HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **CI/CD pipelines** for kernel compilation and automated validation across multiple hardware platforms, reducing manual testing overhead by approximately **40%**.
- Investigated and resolved **kernel regressions** and performance bottlenecks using tools such as **perf** and **ftrace**, ensuring stability for core OS components.
- Collaborated with cross-functional teams to port and maintain **device drivers**, facilitating the integration of new hardware into the Linux kernel.
- Contributed to the development of **low-level system software**, focusing on kernel module stability and efficient resource utilization.
- Managed the **kernel lifecycle** from development through production, including automated testing, debugging, and quality assurance for releases.
- Participated in **kernel tuning** efforts to optimize stability and performance across various subsystems, including memory management and scheduling.
- Containerized kernel development and testing environments using **Docker**, enabling consistent builds and deployments on **AWS EC2** instances.

SKILLS

Kernel Development – Linux Kernel, Kernel Modules, Device Drivers, Low-Level System Software, Kernel Tuning, Kernel Regressions, Kernel Panics, Performance Analysis

Core OS Concepts – Processes, Memory Management, Filesystems, Networking, Scheduling, Interrupts, Hardware/Driver Interactions

Programming Languages – C, Python, Bash, Rust, Go

Debugging & Profiling Tools – perf, ftrace, bpftrace, crash, gdb

CI/CD & Automation – Jenkins, Git, Docker, Automated Validation, CI Pipelines, RPM Packaging

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), RHEL, Linux Environments

PROJECTS

Linux Kernel CI/CD and Validation Framework

- Developed an automated CI/CD system in **Python** and **Bash** to compile, test, and validate Linux kernel builds for various architectures.
- Integrated **Jenkins** pipelines with Git repositories for continuous integration, automatically triggering builds and running test suites on new code commits.
- Implemented detailed logging and reporting mechanisms using **Splunk** to track build status, test results, and identify performance regressions.

Device Driver Debugging and Optimization for Embedded Systems

- Debugged and optimized a custom **C**-based device driver for a sensor array, improving data acquisition latency by **18%**.
- Utilized **ftrace** and kernel debugging tools to pinpoint race conditions and memory leaks within the driver's interrupt handling routines.
- Documented driver behavior, limitations, and performance characteristics for integration into larger embedded Linux systems.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)